SG No. 199 T^5 $I2_13$ [cubic]

* plus set: $+[0,\,0,\,0]$, $+\left[\frac{1}{2},\,\frac{1}{2},\,\frac{1}{2}\right]$
* Wyckoff site: `8a`, site symmetry: `.3.`

Table 1: Wyckoff bond: `8a@8a`, bond symmetry: `.3.`

No.	vector	center	mapping
1	$[X,\,X,\,X]$	$[x,\,x,\,x]$	`[1,5,9]`
2	$[-X,\,-X,\,X]$	$\left[\frac{1}{2}-x,\,-x,\,x+\frac{1}{2}\right]$	`[2,7,12]`
3	$[-X,\,X,\,-X]$	$\left[-x,\,x+\frac{1}{2},\,\frac{1}{2}-x\right]$	`[3,8,10]`
4	$[X,\,-X,\,-X]$	$\left[x+\frac{1}{2},\,\frac{1}{2}-x,\,-x\right]$	`[4,6,11]`

Table 2: Wyckoff bond: `24b@8a`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X,\,Y,\,Z]$	$[x,\,x,\,x]$	`[1]`
2	$[-X,\,-Y,\,Z]$	$\left[\frac{1}{2}-x,\,-x,\,x+\frac{1}{2}\right]$	`[2]`
3	$[-X,\,Y,\,-Z]$	$\left[-x,\,x+\frac{1}{2},\,\frac{1}{2}-x\right]$	`[3]`
4	$[X,\,-Y,\,-Z]$	$\left[x+\frac{1}{2},\,\frac{1}{2}-x,\,-x\right]$	`[4]`
5	$[Z,\,X,\,Y]$	$[x,\,x,\,x]$	`[5]`
6	$[Z,\,-X,\,-Y]$	$\left[x+\frac{1}{2},\,\frac{1}{2}-x,\,-x\right]$	`[6]`
7	$[-Z,\,-X,\,Y]$	$\left[\frac{1}{2}-x,\,-x,\,x+\frac{1}{2}\right]$	`[7]`
8	$[-Z,\,X,\,-Y]$	$\left[-x,\,x+\frac{1}{2},\,\frac{1}{2}-x\right]$	`[8]`
9	$[Y,\,Z,\,X]$	$[x,\,x,\,x]$	`[9]`
10	$[-Y,\,Z,\,-X]$	$\left[-x,\,x+\frac{1}{2},\,\frac{1}{2}-x\right]$	`[10]`
11	$[Y,\,-Z,\,-X]$	$\left[x+\frac{1}{2},\,\frac{1}{2}-x,\,-x\right]$	`[11]`
12	$[-Y,\,-Z,\,X]$	$\left[\frac{1}{2}-x,\,-x,\,x+\frac{1}{2}\right]$	`[12]`

* Wyckoff site: `12b`, site symmetry: `2..`

Table 3: Wyckoff bond: `12a@12b`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[0,\,X,\,Y]$	$\left[x,\,0,\,\frac{1}{4}\right]$	`[1,-4]`
2	$[0,\,-X,\,Y]$	$\left[\frac{1}{2}-x,\,0,\,\frac{3}{4}\right]$	`[2,-3]`
3	$[Y,\,0,\,X]$	$\left[\frac{1}{4},\,x,\,0\right]$	`[5,-8]`
4	$[Y,\,0,\,-X]$	$\left[\frac{3}{4},\,\frac{1}{2}-x,\,0\right]$	`[6,-7]`
5	$[X,\,Y,\,0]$	$\left[0,\,\frac{1}{4},\,x\right]$	`[9,-12]`
6	$[-X,\,Y,\,0]$	$\left[0,\,\frac{3}{4},\,\frac{1}{2}-x\right]$	`[10,-11]`

Table 4: Wyckoff bond: `12b@12b`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[X,\,0,\,0]$	$\left[x,\,0,\,\frac{1}{4}\right]$	`[1,4]`
2	$[-X,\,0,\,0]$	$\left[\frac{1}{2}-x,\,0,\,\frac{3}{4}\right]$	`[2,3]`
3	$[0,\,X,\,0]$	$\left[\frac{1}{4},\,x,\,0\right]$	`[5,8]`
4	$[0,\,-X,\,0]$	$\left[\frac{3}{4},\,\frac{1}{2}-x,\,0\right]$	`[6,7]`

continued ...

Table 4

No.	vector	center	mapping
5	$[0,\, 0,\, X]$	$\left[0,\, \frac{1}{4},\, x\right]$	[9,12]
6	$[0,\, 0,\, -X]$	$\left[0,\, \frac{3}{4},\, \frac{1}{2}-x\right]$	[10,11]

Table 5: Wyckoff bond: 24c@12b, bond symmetry: 1

No.	vector	center	mapping
1	$[X,\, Y,\, Z]$	$\left[x,\, 0,\, \frac{1}{4}\right]$	[1]
2	$[-X,\, -Y,\, Z]$	$\left[\frac{1}{2}-x,\, 0,\, \frac{3}{4}\right]$	[2]
3	$[-X,\, Y,\, -Z]$	$\left[\frac{1}{2}-x,\, 0,\, \frac{3}{4}\right]$	[3]
4	$[X,\, -Y,\, -Z]$	$\left[x,\, 0,\, \frac{1}{4}\right]$	[4]
5	$[Z,\, X,\, Y]$	$\left[\frac{1}{4},\, x,\, 0\right]$	[5]
6	$[Z,\, -X,\, -Y]$	$\left[\frac{3}{4},\, \frac{1}{2}-x,\, 0\right]$	[6]
7	$[-Z,\, -X,\, Y]$	$\left[\frac{3}{4},\, \frac{1}{2}-x,\, 0\right]$	[7]
8	$[-Z,\, X,\, -Y]$	$\left[\frac{1}{4},\, x,\, 0\right]$	[8]
9	$[Y,\, Z,\, X]$	$\left[0,\, \frac{1}{4},\, x\right]$	[9]
10	$[-Y,\, Z,\, -X]$	$\left[0,\, \frac{3}{4},\, \frac{1}{2}-x\right]$	[10]
11	$[Y,\, -Z,\, -X]$	$\left[0,\, \frac{3}{4},\, \frac{1}{2}-x\right]$	[11]
12	$[-Y,\, -Z,\, X]$	$\left[0,\, \frac{1}{4},\, x\right]$	[12]

* Wyckoff site: 24c, site symmetry: 1

Table 6: Wyckoff bond: 24a@24c, bond symmetry: 1

No.	vector	center	mapping
1	$[X,\, Y,\, Z]$	$[x,\, y,\, z]$	[1]
2	$[-X,\, -Y,\, Z]$	$\left[\frac{1}{2}-x,\, -y,\, z+\frac{1}{2}\right]$	[2]
3	$[-X,\, Y,\, -Z]$	$\left[-x,\, y+\frac{1}{2},\, \frac{1}{2}-z\right]$	[3]
4	$[X,\, -Y,\, -Z]$	$\left[x+\frac{1}{2},\, \frac{1}{2}-y,\, -z\right]$	[4]
5	$[Z,\, X,\, Y]$	$[z,\, x,\, y]$	[5]
6	$[Z,\, -X,\, -Y]$	$\left[z+\frac{1}{2},\, \frac{1}{2}-x,\, -y\right]$	[6]
7	$[-Z,\, -X,\, Y]$	$\left[\frac{1}{2}-z,\, -x,\, y+\frac{1}{2}\right]$	[7]
8	$[-Z,\, X,\, -Y]$	$\left[-z,\, x+\frac{1}{2},\, \frac{1}{2}-y\right]$	[8]
9	$[Y,\, Z,\, X]$	$[y,\, z,\, x]$	[9]
10	$[-Y,\, Z,\, -X]$	$\left[-y,\, z+\frac{1}{2},\, \frac{1}{2}-x\right]$	[10]
11	$[Y,\, -Z,\, -X]$	$\left[y+\frac{1}{2},\, \frac{1}{2}-z,\, -x\right]$	[11]
12	$[-Y,\, -Z,\, X]$	$\left[\frac{1}{2}-y,\, -z,\, x+\frac{1}{2}\right]$	[12]